

Medium-Term Determinants of Current Accounts

Replication and Extensions of Chinn and Prasad

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Macroeconomics Presentation

May 2026

Presentation Roadmap

- 1 Introduction
- 2 Presentation of the Paper
- 3 Data Preparation
- 4 Replication of the Results
- 5 Extensions
- 6 Conclusion

Three equivalent identities

Saving–investment lens: $CA = S - I$

Trade lens: $CA = X - M + NII + NUT$

Financial account lens: $CA = -\Delta FA$

- CA summarises a country's external position and the forces shaping saving and investment across borders
- Understanding medium-term CA dynamics is central to sustainability assessments and exchange rate policy
- The 2000s global imbalances, US deficits of 5–6% of GDP, large Asian surpluses, raised fundamental questions about adjustment risks

Research question - Chinn and Prasad (2003)

Which structural fundamentals explain medium-term CA positions across industrial and developing countries over 1971–1995?

$$\left(\frac{CA}{GDP}\right)_{i,t} = \alpha + \beta_1 \text{NFAGDP}_{i,t} + \beta_2 \text{GOVBGDP}_{i,t} + \beta_3 \text{RELY}_{i,t} + \dots + \delta_t + \varepsilon_{i,t}$$

Regressors and motivation:

- NFA/GDP — external adjustment
- Fiscal balance — twin deficits
- Relative income — development stage
- Dependency ratios — life-cycle saving
- Trade openness — capital attractiveness
- Financial depth — investment constraints
- ToT volatility — precautionary saving
- δ_t — global capital flow cycles

Sample

18 industrial + 71 developing countries | Annual data 1971–1995

Key methodological choices:

- **5-year non-overlapping averages** (1971–75, . . . , 1991–95) filters business cycle variation; reduces measurement error in developing country data
- **No country fixed effects** , ~40% of CA variation is cross-sectional; FE would sweep away the cross-country patterns of interest
- **Time dummies** in all panel regressions, captures global capital-flow cycles (Latin debt crisis, 1990s emerging market flows)
- **Separate regressions** for industrial vs. developing countries, slope coefficients differ significantly across groups (formally tested)
- **Oil-exporter dummy**, positive and significant; accounts for structurally different saving behaviour

Key Findings

Initial NFA position

Positive and significant across most specifications. Creditor countries earn investment income; high-liability developing countries attract inflows due to high productivity.

Fiscal balance — partial Ricardian offset

A 1 pp improvement in the budget balance $\Rightarrow$ **+0.40 pp** in CA/GDP for developing countries ($\sim 60\%$ private offset). Coefficient near zero and insignificant for industrial countries.

Developing-country specifics

- **Financial deepening:** positive CA effect, driven by depressed investment
- **ToT volatility:** higher uncertainty $\Rightarrow$ precautionary saving $\Rightarrow$ larger surpluses
- **Trade openness:** negative — open economies attract more foreign capital

Panel $\bar{R}^2 \approx 0.44$ for developing countries.

Core challenge

Original databases discontinued $\Rightarrow$ alternative sources needed, especially pre-1980 coverage

Main sources used:

- **WDI**: CA/GDP, dependency ratios, OPEN, FDEEP, REER, ToT
- **IFS / IMF**: NFA/GDP, GOVBGDP
- **Chinn-Ito index**: capital account openness
- **Penn World Tables**: RELY
- **Bruegel**: improved REER
- **IMF Public Finances**: alternative GOVBGDP
- **WEO**: early NSGDP observations

When one source was insufficient: merge two sources, keeping one per country (least NAs over 1971–1995 wins). Applied to NSGDP, GOVBGDP, and TOTSD.

Replication Strategy

Division of labour

- One member: data pipeline
- Two members: regression code and interpretation, run independently then cross-checked
- Extensions: all three members (separate scripts)

Tools

- **Git** for version control
- **R**: plm, fixest, stargazer
- **Claude** for tedious coding tasks and sourcing data gaps

Key replication divergence

Fiscal balance coefficient **does not replicate** — small, unstable, insignificant throughout. NFA position replicates robustly across all specifications.

Replication: Results

What replicates well

- **NFA** → **CA**: positive and significant across all specs (scaling difference accounted for)
- **CA persistence**: lagged CA coefficient ≈ 0.55 – 0.59 vs. 0.57 in original

New results

- **GDP growth**: negative and significant (-0.245^{**}) — higher growth attracts capital inflows, financing larger deficits
- **REER**: -4.01^{***} in annual data (insignificant in original)
- **Demographics**: both dependency ratios significant (stronger than original)

Variance decomposition

	CP (2003)	Ours
Industrial		
Cross-section	39%	52%
Over time	61%	48%
Developing		
Cross-section	44%	33%
Over time	56%	67%

Reinforces the case for OLS over FE in the industrial country sample.

Extensions: Motivation and Structure

Objective. Assess whether the Chinn–Prasad framework remains valid once the sample is extended beyond 1995 and enriched with crisis-related dynamics.

Three extensions

- 1 **Updated sample:** re-estimate the benchmark model using post-1995 data.
- 2 **Financial crises:** examine current account reversals around major crisis episodes.
- 3 **Saving decomposition:** identify whether fiscal effects operate through private saving or investment.

Guiding question: are current account determinants stable over time, or have global imbalances and crises changed the empirical relationships?

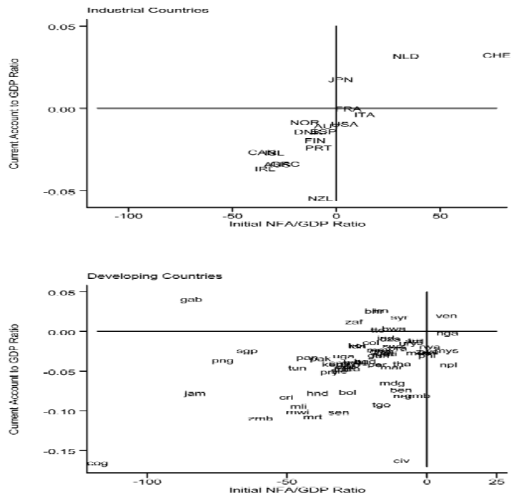
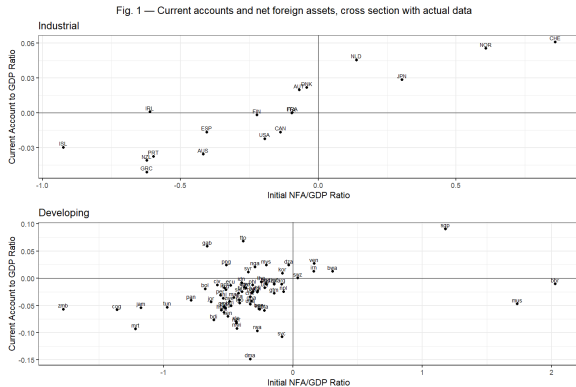


Fig. 1. Current accounts and net foreign assets, cross section.



Panel Estimates: Robust Fiscal Channel

	Full	Excl. Africa	Industrial	Developing	Dev. excl. Africa
OLS with time effects (Table 3)					
Govt. budget balance	0.323***	0.309***	0.447***	0.344***	0.348***
NFA / GDP	1.469	0.934	1.777	0.746	0.095
Two-way fixed effects (Table 4)					
Govt. budget balance	0.340***	0.269***	0.827***	0.329***	0.265***
NFA / GDP	0.643*	0.226	1.114	0.508	0.029

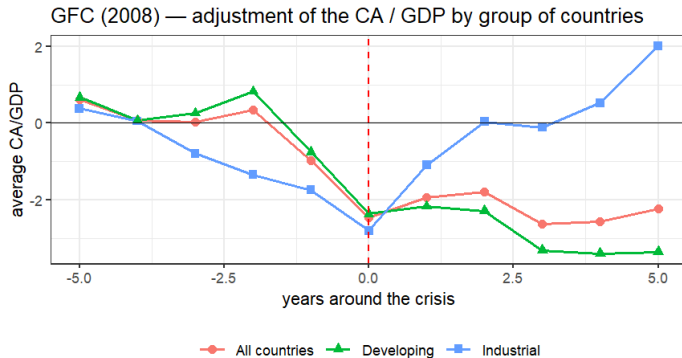
- Fiscal coefficient positive and significant across **all ten specifications** OLS and FE for every country group.
- Industrial country estimate **strengthens** under FE (0.827) relative to original paper (0.141, insignificant), driven by post-2008 fiscal heterogeneity.
- NFA weakens under fixed effects, consistent with its predominantly cross-sectional variation documented in Table 1.
- Twin-deficit relationship **has strengthened** relative to the original 1971–1995 sample.

Annual Data: Persistence and Exchange Rate Adjustment

	Full	Excl. Africa	Industrial	Developing	Dev. excl. Africa
Govt. budget balance	0.197***	0.105***	0.212***	0.225***	0.133***
Lagged CA/GDP	0.711***	0.749***	0.762***	0.686***	0.705***
Lagged $\Delta \log$ REER	-3.041***	-2.859***	-7.907***	-2.685***	-2.521***
Financial deepening	0.009**	0.006**	0.007	0.011*	0.012**
R^2	0.661	0.704	0.871	0.624	0.660

- **Persistence:** lagged CA coefficient between 0.69 and 0.76 — strong but incomplete inertia.
- **Fiscal effect:** smaller than in 5-year specs (0.197 vs 0.323) — the lagged DV absorbs dynamic adjustment and cyclical noise.
- **REER:** a 1% lagged appreciation deteriorates the CA by 2.5 to 7.9 pp — largest for industrial countries.
- **Financial deepening:** positive at annual frequency, contrasting with negative medium-term sign — reflects structural investment channel, not short-run behavior.

Global Financial Crisis: Different Adjustment Pattern



- All groups reach a trough around 2008.
- Industrial countries then experience a sustained current account improvement.
- Developing countries remain in persistent deficit after the shock.

Implication: the GFC should not be treated as a symmetric crisis shock across country groups.

Extension: Gross Flows

Motivation: gross flows (inflows + outflows) can mask dynamics invisible to CA analysis, i.e. simultaneous collapse of both leaves CA unchanged.

	Full	Excl. Africa	Industrial	Developing	Dev. excl. Africa
Govt. budget balance	0.175	0.242	0.138	0.342	0.370
NFA / GDP	-0.691	-1.374	-31.189***	2.494***	2.254***
Relative income ²	35.691***	40.060***	139.903	66.153***	68.053***
Openness	0.115***	0.131***	0.531***	0.062*	0.075**
R^2	0.354	0.367	0.698	0.473	0.486

- Structural fundamentals explain little; **openness is the dominant driver**
- High NFA $\Rightarrow$ fewer gross flows for industrial countries only
- Fiscal balance: **not significant** for gross flows

Gross Flows: Inflows vs. Outflows

Inflows

- Higher fiscal balance $\Rightarrow$ **fewer** inflows (-0.200^{**}): lower financing need reduces foreign capital demand
- Higher growth $\Rightarrow$ **more** inflows ($+0.423^{***}$): domestic capacity attracts foreign investors
- Capital controls *increase* inflows (not a very robust result)

Outflows

- Mirror the total gross flow picture
- Openness dominant predictor
- Relative income quadratic significant for developing countries

Takeaway

Fiscal and growth fundamentals operate through the **financing side** (inflows), not outflows

Extension: Logit Approach to the Asian Crisis

Setup: Asian crisis (1997) affects only a subset of countries $\Rightarrow$ binary model appropriate

- Dependent variable: dummy = 1 for the **3 years preceding** the crisis for affected countries (TH, KR, ID, MY, PH)
- Logit with year fixed effects
- Pseudo- $R^2 = 0.671$

Variable	Coeff.	Std. Err.
Gross inflows / GDP	0.554***	(0.193)
Relative income	-16.54**	(6.940)
Govt. balance	0.956**	(0.384)
National saving / GDP	0.450**	(0.176)
Capital account openness	3.951*	(2.154)
Financial deepening	-0.061*	(0.035)

Surprising result

If taken at face value, high fiscal surpluses and high savings **increase** crisis probability

Saving Decomposition and Ricardian Offset

Accounting identity:

$$CA = NS - I$$

Empirical implication:

$$\beta^I = \beta^{NS} - \beta^{CA}$$

- In the updated sample, the investment response is small:

$$0.393 - 0.323 = 0.070$$

- Excluding Africa, it is also negligible:

$$0.232 - 0.309 = -0.077$$

- Therefore, the fiscal-current account channel mainly operates through saving.

Extension

We separate public and private saving to test the strength of the Ricardian offset.

Ricardian Offset: Regression Results

	Full	Excl. Africa	Industrial	Developing	Dev. excl. Africa
Govt. balance	-0.625***	-1.065***	0.087	-0.754***	-1.140***
Financial deepening	-0.0004	-0.003	-0.065**	-0.0003	0.080***
GOV $\times$ FDEEP	0.0003	0.006	-0.007	0.0002	0.006*
Observations	521	367	76	445	291
R^2	0.321	0.348	0.881	0.377	0.434

- Private saving falls strongly when the fiscal balance improves.
- The offset is large in developing countries, especially outside Africa.
- **Financial deepening does not amplify the Ricardian offset** — the total marginal effect is essentially flat across the FDEEP distribution.
- Suggests precautionary saving motives, not formal credit access, drive the behavioral response.

Extensions: Main Findings

- ① **Stability of core determinants:** fiscal balance and NFA remain central predictors of current accounts.
- ② **Stronger fiscal channel:** the twin-deficit relationship is more robust in the updated sample.
- ③ **Crisis asymmetry:** Asian and Argentine crises generate sharp reversals, while the GFC has heterogeneous effects.
- ④ **Saving channel dominance:** fiscal shocks affect current accounts mostly through private saving, not investment.
- ⑤ **Limited role of financial depth:** formal financial deepening does not explain the strength of the Ricardian offset.

Bottom line

The Chinn–Prasad framework remains useful, but post-1995 global imbalances and crisis episodes require dynamic and crisis-specific extensions.

